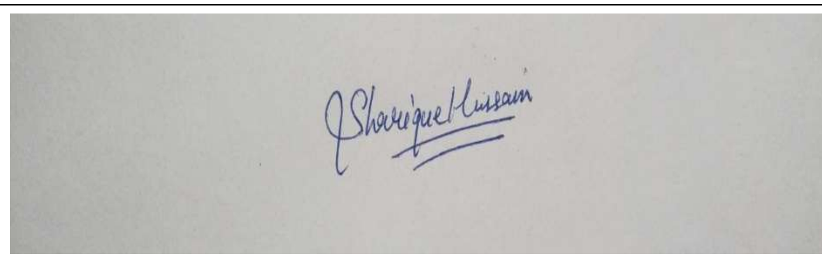
<u>**Annexure3b- Complete filing**</u>

<u>**INVENTION DISCLOSURE FORM**</u>

Details of Invention for better understanding:

1. **TITLE:** Real-Time Context-Aware Multimodal Sign Language to Emotion-Preserved Audio Conversion System

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

A. Full name	Md Sharique Hussain
Mobile Number	9279188328
Email (personal)	shariquehussain03@gmail.com
UID/Registration number	12308048
Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
Signature **(Mandatory)**	

EXTERNAL INVENTOR(S): (INVENTORS NOT WORKING IN LPU)

***For External Inventors*, NOC (No Objection Certificate) from the affiliated institute/university/Industry/lab etc. is mandatory for each individual inventor and their respective topic. For NOC, format is attached below.**

DESCRIPTION OF THE INVENTION: The present invention relates to an artificial intelligence-based communication system that converts sign language gestures into natural, emotion-preserved spoken audio in real time.

The system integrates:

- Hand gesture recognition
- Facial emotion detection
- Motion intensity analysis
- Context-aware language reconstruction
- Emotion-conditioned neural speech synthesis

Unlike traditional systems that perform linear gesture-to-text-to-speech translation, this invention employs a parallel weighted multimodal fusion architecture to preserve intent and emotional tone during translation.

The system is optimized for real-time performance and can operate on edge devices for low-latency accessibility applications.

A. **PROBLEM ADDRESSED BY THE INVENTION:** Existing sign language translation systems suffer from:

1. Loss of emotional expression during translation.
2. Literal word mapping without contextual correction.
3. Robotic or flat speech output.
4. Dependence on cloud processing.
5. Poor real-time responsiveness.

Sign language inherently contains emotional and expressive elements through facial expressions, gesture intensity, and posture. Current systems ignore these parameters.

This invention addresses the research gap by introducing an integrated multimodal processing framework that preserves emotional intent and conversational naturalness.

B. **OBJECTIVE OF THE INVENTION (Provide minimum two)**

Objective 1:
To develop a real-time multimodal sign language recognition system integrating gesture, facial emotion, and motion intensity analysis for accurate intent classification.

Objective 2:
To design an emotion-conditioned speech synthesis engine that produces natural, contextually reconstructed spoken audio from sign input.

Objective 3:
To implement a weighted parallel data fusion architecture that enhances translation accuracy and preserves expressive intent.

Objective 4:
To optimize the system for low-latency edge-device deployment to ensure accessibility in low-connectivity environments.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY: Describe your invention fulfil the research gap?

Sr. No.	Patent I'd	Abstract	Research Gap	Novelty
1.	US20180012345A1	A vision-based sign language recognition system that captures hand gestures using a camera and converts them into text and optionally into speech using machine learning models such as CNN and LSTM. The system follows a gesture-to-text-to-speech pipeline.	The system performs linear translation and does not incorporate facial emotion detection, motion intensity analysis, or intent-based semantic reconstruction. Emotional nuance and contextual grammar restoration are absent. Speech output is neutral and non-adaptive.	The present invention introduces a parallel weighted multimodal fusion architecture integrating gesture, facial emotion, and motion intensity. It performs intent-based classification and emotion-conditioned neural speech synthesis, preserving expressive tone and contextual meaning.

2.	IN202041012345A	An image-processing-based sign language converter that detects predefined hand gestures and maps them to a stored word database, generating corresponding synthetic speech output.	The system relies on predefined gesture-word mapping and lacks dynamic AI-based interpretation. It does not support multimodal emotion detection, contextual language reconstruction, or adaptive speech personalization. Processing architecture is sequential and database-dependent.	The present invention uses AI-driven intent modeling, contextual grammar reconstruction, and emotion-preserving speech synthesis. It employs a real-time multimodal processing engine optimized for edge deployment, enabling expressive and natural communication rather than static word mapping.

D. DETAILED DESCRIPTION:

The present invention relates to a real-time, context-aware, multimodal sign language to emotion-preserved audio conversion system designed to translate visual sign inputs into natural, emotionally expressive speech output.

The invention integrates computer vision, machine learning, multimodal data fusion, natural language processing, and neural speech synthesis within a unified processing architecture.

1. System Overview

The system comprises the following primary modules:

1. Input Capture Module

2. Parallel Multimodal Processing Engine
3. Weighted Multimodal Fusion Layer
4. Intent Classification Module
5. Context Restoration Engine
6. Emotion-Conditioned Speech Synthesis Engine
7. Natural Audio Output Interface

Each module operates sequentially in data flow while internally supporting parallel computation for low-latency real-time processing.

2. Input Capture Module

The Input Capture Module includes:

- A camera device configured to capture real-time video frames of a user performing sign language.
- A hand landmark detection submodule configured to extract skeletal hand keypoints and spatial coordinates.
- A facial emotion detection submodule configured to analyze facial muscle movements, eyebrow tension, lip curvature, and eye openness.
- A motion intensity analysis submodule configured to measure gesture velocity, acceleration, and amplitude variations.

The module outputs structured multimodal data including gesture coordinates, facial expression vectors, and motion intensity parameters.

3. Parallel Multimodal Processing Engine

The Parallel Multimodal Processing Engine processes extracted features simultaneously through:

- A Gesture Recognition Model trained to classify sign patterns.
- An Emotion Classification Model trained to detect emotional states such as neutral, happy, angry, sad, urgent, or stressed.
- A Motion Velocity Analyzer that quantifies dynamic movement patterns for intensity-based interpretation.

Unlike linear processing architectures, the present invention performs simultaneous inference across modalities to preserve expressive context.

4. Weighted Multimodal Fusion Layer

The Weighted Multimodal Fusion Layer integrates outputs from gesture recognition, emotion classification, and motion intensity analysis.

This layer performs:

- Parallel data integration
- Confidence scoring of each modality
- Dynamic weighting adjustment based on prediction reliability

The fusion mechanism computes an aggregated multimodal representation that reflects both semantic meaning and emotional intent.

This architecture differs from traditional sequential pipelines by incorporating confidence-based adaptive weighting.

5. Intent Classification Module

The Intent Classification Module receives the fused multimodal representation and performs semantic interpretation.

Instead of mapping gestures directly to predefined words, this module predicts communicative intent using contextual probability modeling.

The module determines:

- Sentence-level meaning
- Speaker emotional state
- Urgency or emphasis

This stage transforms literal gesture recognition into context-aware interpretation.

6. Context Restoration Engine

Sign languages frequently omit grammatical connectors such as auxiliary verbs, articles, or tense markers.

The Context Restoration Engine reconstructs grammatically complete sentences by:

- Predicting omitted connectors
- Adjusting word order where required
- Ensuring linguistic consistency with the target spoken language

This module applies natural language processing techniques to generate structurally accurate spoken sentences without altering intended meaning.

7. Emotion-Conditioned Speech Synthesis Engine

The Emotion-Conditioned Speech Synthesis Engine converts the reconstructed sentence into natural speech output.

The engine:

- Matches tone and pitch according to detected emotional state
- Adjusts speech tempo based on motion intensity
- Supports personalized voice profiles for user identity preservation

Unlike conventional text-to-speech systems that produce neutral output, this module conditions speech synthesis on emotional parameters derived from multimodal analysis.

8. Natural Audio Output

The final processed speech signal is transmitted through an audio output device in real time.

The system is optimized to maintain low latency suitable for live conversational communication.

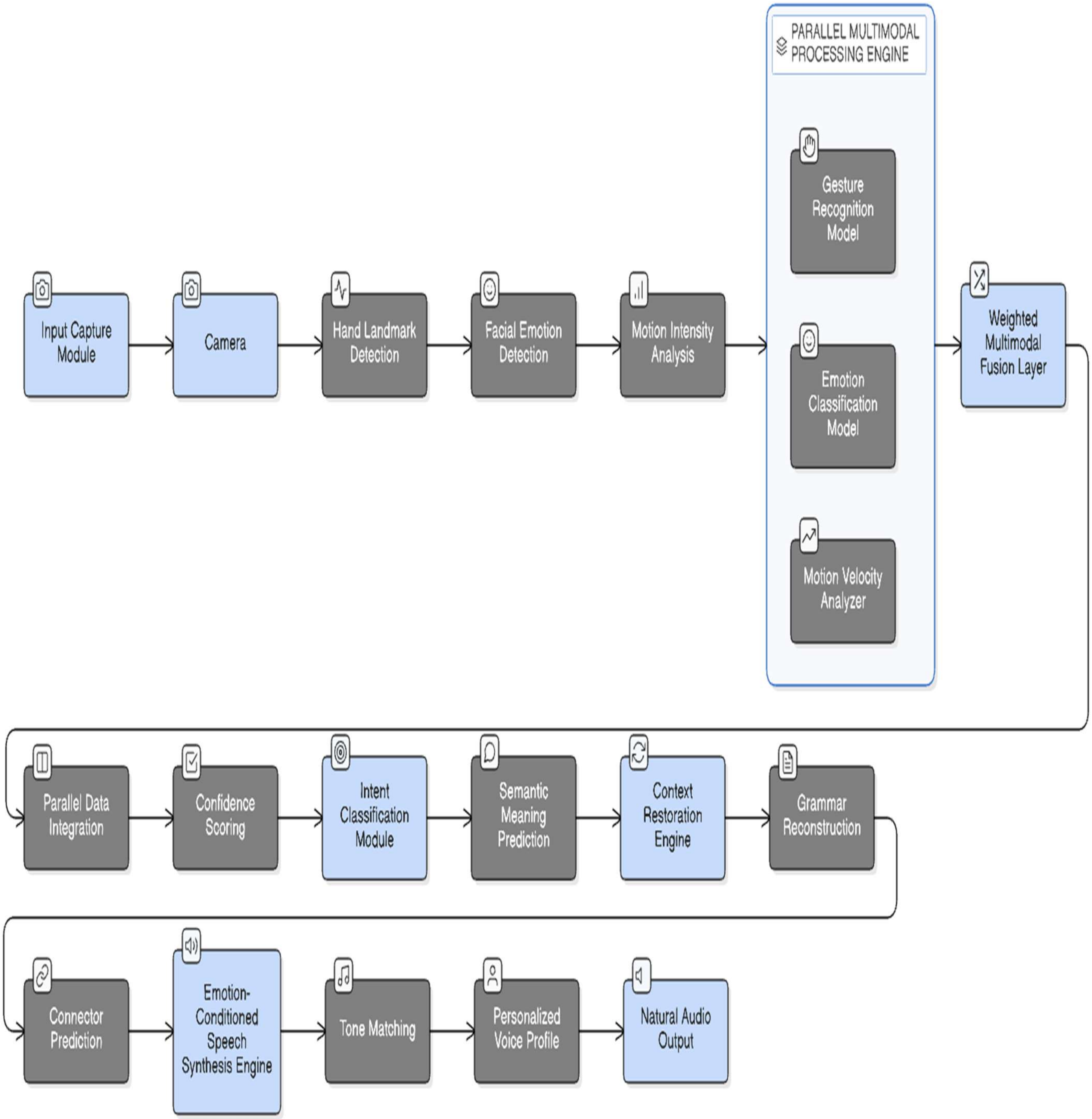
Edge-based deployment may be implemented to reduce dependency on internet connectivity.

9. Technical Advantages

The invention provides the following improvements over prior systems:

1. Emotional preservation during translation.
2. Parallel multimodal fusion architecture.
3. Intent-based semantic interpretation.
4. Context-aware grammatical reconstruction.
5. Emotion-conditioned neural speech generation.
6. Real-time low-latency operation.
7. Edge-device compatibility.

This architecture ensures expressive, natural, and contextually accurate communication between sign language users and non-sign language users.


Input Capture Module
Camera
Hand Landmark Detection
Facial Emotion Detection
Motion Intensity Analysis
PARALLEL MULTIMODAL PROCESSING ENGINE
Gesture Recognition Model
Emotion Classification Model
Motion Velocity Analyzer
Weighted Multimodal Fusion Layer
Parallel Data Integration
Confidence Scoring
Intent Classification Module
Semantic Meaning Prediction
Context Restoration Engine
Grammar Reconstruction
Connector Prediction
Emotion-Conditioned Speech Synthesis Engine
Tone Matching
Personalized Voice Profile
Natural Audio Output

E. RESULTS AND ADVANTAGES:

The present invention provides significant technical and functional improvements over existing sign language translation systems. The detailed results and advantages are as follows:

1. Preservation of Emotional Expression

Unlike conventional systems that convert gestures into neutral speech, the present invention preserves emotional context by integrating facial emotion detection and motion intensity analysis into the translation pipeline. This results in speech output that reflects the speaker's emotional state, ensuring expressive and authentic communication.

2. Improved Translation Accuracy through Multimodal Fusion

The weighted multimodal fusion architecture integrates gesture recognition, emotion classification, and motion analysis simultaneously. By assigning confidence scores and dynamically adjusting modality weights, the system improves semantic accuracy and reduces misclassification compared to single-modality systems.

3. Intent-Based Interpretation Instead of Literal Mapping

Traditional systems perform direct gesture-to-word mapping. The present invention introduces an intent classification module that predicts communicative meaning at the sentence level. This reduces ambiguity and enhances contextual understanding.

4. Context-Aware Grammar Reconstruction

Sign languages often omit auxiliary verbs, articles, and tense markers. The context restoration engine reconstructs grammatically complete spoken sentences without altering intended meaning. This ensures that output speech is structurally correct and conversationally natural.

5. Emotion-Conditioned Neural Speech Synthesis

The speech synthesis engine adjusts tone, pitch, tempo, and vocal energy according to detected emotional parameters. This eliminates robotic output and enhances conversational realism.

6. Personalized Voice Profile Support

The system supports user-defined voice characteristics, enabling individuals to select or train voice profiles that represent their identity. This ensures personal expression and avoids generic synthetic output.

7. Real-Time Low-Latency Performance

The architecture supports parallel processing and optimized inference mechanisms, enabling real-time operation suitable for live conversation scenarios such as classrooms, hospitals, and public offices.

8. Edge-Device Deployment Capability

The system is optimized for deployment on embedded and edge devices, reducing dependency on cloud-based infrastructure. This allows usage in low-connectivity or rural environments.

9. Reduced Communication Barriers in Critical Environments

By preserving emotional and contextual information, the invention enhances clarity in high-stakes communication scenarios such as emergency services, healthcare consultations, and legal proceedings.

10. Scalability Across Multiple Sign Languages

The modular architecture allows adaptation to different regional sign languages through retraining of recognition models without altering the core processing pipeline.

11. Increased Robustness Against Ambiguity

The confidence-weighted multimodal fusion mechanism reduces errors caused by partial occlusion, lighting variation, or inconsistent gesture speed.

12. Foundation for Bidirectional Communication Systems

The architecture can be extended to integrate speech-to-sign avatar systems, creating a complete two-way accessibility communication framework.

Summary of Technical Impact

The invention does not merely translate sign language into speech; it reconstructs expressive, contextually accurate, and emotionally aligned communication using a novel multimodal fusion architecture and emotion-conditioned synthesis pipeline.

F. EXPANSION: Any variables which are necessary for your invention to be covered?

NA

G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION: Is your working prototype or other ready. Provide the images/data of the prototype. If no, how much time is required for the same?

Prototype Status:
Under Development.

Estimated Time Required:
4–6 months for initial functional prototype including gesture recognition, emotion detection, and basic speech output integration.

G. EXISTING DATA: Any clinical or comparative data necessary enough to support your invention. (Comparative)

NA

4. **USE AND DISCLOSURE (IMPORTANT):** Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?	YES ( )	NO (X)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ( )	NO (X)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ( )	NO (X)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ( )	NO (X)
E. Name of Regulatory body or any other approvals if required.	YES ( )	NO (X)

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us.

NA

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity).

NA

7. Potential Chances of Commercialization.

Yes

8. List of companies which can be contacted for commercialization along with the website link.

Provide the details of companies.

9. Any basic patent which has been used and we need to pay royalty to them.

NA

10. **FILING OPTIONS:** Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings (Mandatory to mention).

Complete filing

11. **KEYWORDS:** Please provide right keywords for searching your invention.

Sign Language Recognition

Multimodal Fusion

Emotion-Preserved Translation

Neural Speech Synthesis

Accessibility AI

Real-Time Gesture Recognition

Intent Classification

Edge AI Communication System